

FAUZAN FEBRIANSYAH

Embedded System Engineer – Gamantaray UGM

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SUMMARY

Fauzan is an undergraduate student currently pursuing a degree in Internet Engineering Technology. He is active in student organizations and multidisciplinary teams, with a strong interest in technology planning and implementation. His focus lies in digital infrastructure and IoT development, particularly networking, embedded systems, and system architecture. As a highly adaptable and curious learner, Fauzan is currently focused on developing reliable control and visualization systems for autonomous robotics (AUV/ROV) projects at Gamantaray UGM.

EDUCATION

Bachelor of Applied Science (D4) in Internet Engineering Technology

July 2024 – Present

Universitas Gadjah Mada – Sleman, Indonesia

Cumulative GPA : 3.87/4.00

EXPERIENCE

Laboratory Teaching Assistant

August 2026 – Present

Vocational College, Universitas Gadjah Mada - Yogyakarta, Indonesia

- Instructed and mentored 50 students in programming tools and basic electronics concepts using an interactive approach and real-world case studies.
- Managed weekly grading and evaluations for 50+ technical reports across 2 classes, providing constructive feedback to improve students' data analysis quality.
- Minimized technical disruptions during lab sessions by offering real-time instrument troubleshooting support and conducting preventive equipment checks.
- Coordinated weekly lab workflows with course professors to ensure all intended learning outcomes were achieved on schedule.

Event Operational Manager

January 2026 – Present

Universitas Gadjah Mada Block Chain Club - Yogyakarta, Indonesia

- Led the planning and execution of various blockchain event formats, ranging from seminars to workshops, ensuring all activities aligned with target objectives.
- Managed and coordinated 3 events with a total of 300+ attendees.
- Directed the operational team to ensure efficient task delegation and optimal workflow execution across all event projects.

Embedded System Engineer

December 2025 – Present

Gamantaray - Yogyakarta, Indonesia

- Built a 20 Hz onboard control loop integrating GCS commands, STM32 sensor data, Pixhawk depth telemetry, vision status, and failsafe logic.
- Implemented 115,200-baud STM32 serial communication across 5 direct hardware signals, such as encoder ticks, target position, PWM output, ballast motor speed, and gripper status.
- Integrated Pixhawk via MAVLink for controlling vehicle degrees of freedom (pitch, roll, yaw, and surge) with RC PWM commands bounded to 1100–1900 μ s.
- Constructed a bidirectional UDP network across 4 ports: command receiving (5005), telemetry transmission (5006), and 2 video streaming channels (5000/5001).
- Designed and deployed 3 PID control loops for depth/ballast, pitch stabilization, and heading hold, complete with output clamping and integral anti-windup protection.

PROJECTS

Voyager Ship - Semi-Autonomous Marine Trash Collector Vessel

May 2026 – July 2026

- Programmed Arduino microcontrollers to generate PWM signals for controlling 2 ESC units, ensuring accurate maneuvering response from dual thrusters.
- Executed electrical wiring linking microcontrollers to ESCs and safely distributed direct power supply to both thrusters.
- Performed structural troubleshooting by identifying and sealing hull leaks to safeguard internal electronics.

Safara Gate - PPE Compliance Monitoring System via Computer Vision

April 2026 – June 2026

- Developed object detection models using YOLO and Roboflow by collecting and annotating a custom dataset of 11,000+ images across 7 classes to achieve high recognition accuracy.
- Designed circuit layouts on perfboards and soldered step-down power modules to ensure stable 5V and 12V voltage distribution across system components.
- Integrated ESP32 microcontrollers, RFID modules, and I2C LCD screens into a compact autonomous security system housed inside a tubular enclosure.
- Deployed a 2-worker-node Kubernetes cluster to host the management dashboard server, enabling high-availability real-time access monitoring.

Hydrovia - Smart Pipe Water Leakage Monitoring System

October 2025 – December 2025

- Programmed an ESP32 microcontroller as an IoT gateway integrating 2 flow sensors, an HC-SR04 ultrasonic water tank sensor, and automated control for a water pump actuator.
- Designed electrical schematics and wiring to safely distribute power (12V for the pump actuator and 5V for microcontroller and sensors).
- Configured MQTT telemetry data pipelines to stream real-time sensor metrics to the server at 1-second intervals.
- Built an interactive Node-RED Smart Water Monitoring dashboard to visualize daily volume metrics, display real-time status indicators, and provide remote valve override and data reset controls.

TRAINING & CERTIFICATIONS

Network Design Training

February 2026 – July 2026

- Analyzed business and technical requirements to define system specifications.
- Characterized network behavior and selected appropriate hardware components.
- Determined optimal switching and routing protocols tailored to operational needs.
- Designed logical and physical network topologies.
- Prepared comprehensive documentation and performed network design evaluations.

MikroTik MTCNA Training

August 2025 – December 2025

- Configured routers and Internet gateway connectivity.
- Managed network bandwidth allocation and routing policies.
- Implemented network security protocols and NAT configurations.
- Established network redundancy and VPN tunneling connections.
- Designed VLAN architectures and network segregation schemes.

MikroTik MTCWE Training

August 2025 – December 2025

- Configured wireless links and monitored RF signal quality metrics.
- Managed access control and Virtual Access Point (VAP) setups.
- Enforced wireless network protection and security standards.
- Deployed Wireless Distribution System (WDS) and Mesh topologies.
- Applied Wi-Fi standards and Quality of Service (QoS) optimization.
- Executed wireless bridging and tunneling solutions.

ACTIVITIES & VOLUNTEERING

Event Staff

September 2025 – February 2026

National Networking Competition - Yogyakarta, Indonesia

- Designed and executed seminar and online class concepts for 600+ participants across Indonesia.
- Coordinated inter-divisional teams during the seminar to ensure smooth event flow and achievement of objectives.
- Prepared human resources (MC, moderators) and technical elements (room layouts, rundowns).
- Adapted and adjusted live operations in real-time to handle unscheduled contingencies effectively.

Event Staff

March 2025 – August 2025

PIONIR Gadjah Mada - Yogyakarta, Indonesia

- Designed and executed opening and closing ceremony concepts for PIONIR Gadjah Mada 2025, serving 10,000+ incoming freshmen.
- Organized talent auditions for new students performing at the closing ceremony and structured performance routines.
- Co-designed interactive web games with the IT division for the official PIONIR website and optimized event support workflows.
- Liaised with the Faculty of Veterinary Medicine and led cross-functional teams to design technical schedules and learning sessions for freshmen.

SKILLS

- **Programming & Tools** : C/C++, Python, YOLO, OpenCV, Roboflow, Github
- **Embedded Systems & Hardware** : STM32, ESP32, Pixhawk (MAVLink), Microcontroller Interfacing, PID Control, Sensor Integration
- **Networking & Cloud Infrastructure** : Mikrotik (Routing/Switching, MTCNA & MTCWE concept), Docker, Kubernetes, Linux (Ubuntu), MQTT, UDP/TCP Protocol